

Thermal performance and optimum concentration of different nanofluids in immersion cooling in data center servers

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ABSTRACT

The application of nanofluids in heat transfer systems is widely recognized for enhancing thermal efficiency compared to conventional fluids, representing a pivotal advancement in energy-saving strategies across a range of commercial applications. However, the selection of nanofluid concentration often lacks a systematic approach, with decisions made independently of specific application requirements, type of nanofluid, cost considerations, and economic factors such as energy cost and system lifetime. This article explores the optimization of nanoparticle concentrations in the immersion cooling system of a data center server, utilizing three distinct nanofluids (Al₂O₃-water, TiO₂-water, and CuO-water). The Iceotope immersion cooling designs were examined across various flow conditions. The investigation involved calculating heat transfer rates and pressure drops through the server by simulating a model using Computational Fluid Dynamics (CFD), Finite Element (FE), and COMSOL Multiphysics® Modeling software for varying volume fractions of nanoparticles. The study proposes a novel methodology to assess potential economic trade-offs associated with nanofluid utilization in immersed liquid-cooled data centers. This approach aims to identify the lowest overall cost within the data center and the maximum viable volume fraction of a nanofluid. Results indicate that Al₂O₃-water nanofluid exhibits superior thermal efficiency due to its higher thermal conductivity, while TiO₂-water and CuO-water are evaluated for comparative performance. However, it also presents challenges, such as higher viscosity leading to increased pressure drop and pumping costs. Economic considerations reveal that, despite its thermal advantages, the most cost-effective fluid is the conventional fluid (pure water). The server incurs a cost of \$706.15. These findings underscore the importance of considering both thermal performance and economic factors when determining the optimal nanofluid concentration for practical application.

1. Introduction

In today's technology-driven world, Information and Communication Technology (ICT) underpins critical sectors such as industry, education, transportation, and economics, with data centers serving as essential infrastructure for storing, processing, and managing vast quantities of data [1,2]. Despite their importance, data centers are highly energy-intensive, with cooling systems accounting for over half of their total energy consumption due to the significant heat generated by central processing units (CPUs). In 2010, Google's data centers alone operated approximately 900,000 servers, consuming 1.9 billion kWh of electricity, and by 2016, global data center energy consumption reached 200 TWh, with projections estimating an increase to 2967 TWh by 2030

[3,4]. As energy costs rise and data centers continue to expand, the imperative to reduce energy consumption, particularly from cooling systems, has become a priority. Effective thermal management is crucial to optimizing CPU performance, extending server lifespan, and minimizing energy use [5]. While traditional air-cooling methods remain widely used, they are often inadequate for handling the thermal loads produced by modern data centers. Liquid cooling systems have emerged as a more effective alternative, offering superior heat absorption and dissipation capabilities, which mitigate hotspots and improve overall thermal performance [6,7]. However, liquid cooling solutions face limitations in heat transfer efficiency due to the lower thermal conductivity of fluids such as water. To address this challenge, the use of nanofluids, base fluids infused with nanoparticles, has been proposed as

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a novel solution to enhance thermal conductivity and heat transfer rates. This approach represents a significant advancement in the design of cooling systems, with nanofluids demonstrating potential for improving energy efficiency in data centers and other applications. Variations in nanofluids, based on the composition, size, shape, and concentration of nanoparticles, offer promising opportunities for further optimizing energy-saving efforts within data centers [8,9]. A range of studies has explored thermal management in nanofluid systems with various nanoparticles and complex flow conditions. For instance, Hayat et al. [10] examined ternary-hybrid nanofluids, showing that variables such as the Forchheimer number significantly affect velocity and temperature distributions under stretching surfaces, with entropy generation increasing under certain thermal parameters. Similarly, Naz et al. [11] analysed the impacts of magneto hydrodynamics and nonlinear convection in porous media, noting that thermal transport rates and entropy generation vary widely with parameters like the Brinkman number and thermal conductivity. These studies highlight the importance of nanoparticle selection and flow dynamics for optimizing heat transfer efficiency across engineering applications.

In recent years, nanofluids have garnered significant attention for their potential to enhance thermal performance in various practical applications, including electronics cooling and heat exchange systems. Researchers have frequently employed the Nusselt number (Nu) and Reynolds number (Re) to evaluate the thermal efficiency of nanofluids. However, this approach can be imprecise since the properties of nanofluids depend on both Nu and Re, and their increased viscosity requires greater pumping power to achieve equivalent Re values as compared to traditional fluids like water. This additional pumping power, which escalates with the concentration of nanoparticles, introduces practical limitations, including higher pressure drops and energy costs, that necessitate careful economic analysis [12–16]. Therefore, the use of nanofluids in practical applications should be subjected to an economic analysis to estimate this trade-off.

In practical applications, nanofluids have shown promise in enhancing heat transfer performance. For instance, Al_2O_3 nanofluids have been demonstrated to reduce CPU operating temperatures by up to 20 %, compared to traditional cooling fluids, while TiO_2 and CuO nanofluids have achieved similar improvements in thermal efficiency, particularly when applied to CPU cooling and microchip heat sinks [17–21]. Although these fluids enhance heat dissipation by lowering surface temperatures, they also incur the drawback of increased pressure losses, which rise with nanoparticle concentration. This trade-off highlights the importance of evaluating both thermal and hydraulic performance, alongside factors such as cost, toxicity, and corrosion, to determine the overall feasibility of nanofluid use in real-world applications [20–22].

Based on the available literature, it seems that nanofluids are promising in terms of enhancing the heat transfer performance. However, nanofluids have been used by some researchers for electronic device cooling, particularly the CPUs. Zahid et al. [23,24] explored $\alpha\text{RT-70HC}$ and $\text{RT54HC}/\text{Al}_2\text{O}_3$ -based nano-enhanced phase change materials (NePCMs) for passive cooling in electronic equipment, testing various heat sink designs. Their results indicated that the unfinned heat sink had limited effectiveness, while circular pin fin and metal foam heat sinks (MFHS) showed significant improvements. The MFHS reduced the base temperature by up to 36.2 % at 0.25 wt% Al_2O_3 nanoparticles under a heat flux of 0.98 kW/m^2 and increased functional time by 400 % at a set temperature of 55°C , outperforming other configurations in thermal management. Roberts and Walker [18] used 20–30 nm-sized Al_2O_3 nanofluids as a coolant in liquid cooling systems for the CPUs. The findings showed a 20 % increase in heat transfer convection compared to deionized (DI) water. Three nanofluids called water / ethylene (base fluid) with Al_2O_3 , SiO_2 and TiO_2 (nanoparticles) have been tested [19] to investigate computer microchip cooling thermal performance. Studies have shown that nanofluids significantly improve heat transfer by reducing the processor's operating temperature compared to base fluids.

Specifically, Al_2O_3 nanofluid achieves an operating temperature that is 5.5°C lower than the base fluid at the same flow rate and a volume fraction of 1.0 vol.%. Economic analysis and power consumption of the various nanofluid concentrations and flow rates were therefore investigated [19]. To investigate the heat transfer rate on a CPU cooling device, two separate nanofluids called Al_2O_3 and carbon nanotubes (CNT) were applied [20]. Their result revealed that, relative to the base fluid for Al_2O_3 nanofluid and CNT nanofluid, the CPU temperature is lower by 20 % and 22 %, respectively. They also suggested using the system's nanofluid for economic matters. In addition, TiO_2 and Cu nanofluids have been applied to evaluate the cooling efficiency for the PC processor of heat sink fans [21]. Their analysis indicates that using nanofluids with a volume fraction of 0.1 % reduces the average processor temperature by approximately 30°C compared to the base fluid at the same flow rate, with this reduction increasing as the volume fraction is increased. They claimed that the nanofluids should not only be selected based on the thermal performance; they should also include nanofluid cost, toxicity and corrosion. An extensive investigation of utilizing nanofluid in micro-channel heat-sinks for cooling electronic chip CPUs has been conducted [25–31]. Research has demonstrated that the use of nanofluids enhances thermal dissipation from the heat source. Moreover, increasing the concentration of nanoparticles in the nanofluid results in a further reduction in the surface temperature of the CPU chip. Nevertheless, this improvement is accompanied by the cost of increased pumping power due to a rise in pressure loss as the ratio of nanoparticles increases. When employing nanofluids, it should be essential to conduct an economic analysis to determine their impact on thermal and hydraulic performance. It has been noticed that when the concentration of nanofluid increases, both heat transmission and pressure drop also rise. Thus, in certain scenarios, nanofluids may only offer economic advantages when the pump cost is low and there is a need for high-density cooling. In addition to the significant expense of nanofluids, they also contribute to higher energy costs and are considered a major obstacle in certain applications of nanofluids. There are a limited number of studies that utilize nanofluids for economic analysis. Faizal et al. [32] conducted an economic analysis on the use of solar collectors with different nanofluids to achieve consistent thermal performance across varying temperatures. Their findings indicate that incorporating nanofluids such as CuO , SiO_2 , TiO_2 , and Al_2O_3 can significantly reduce the required size of the solar collector. Specifically, the reductions are estimated at 26 %, 22 %, 22 %, and 21 %, respectively, for these nanofluids. This reduction in size directly contributes to a decrease in both the weight and manufacturing cost of the solar collectors. Recent studies have shown that nanofluids, such as Al_2O_3 -water and TiO_2 -water, can significantly improve cooling efficiency in high-density electronic applications by enhancing thermal conductivity and reducing CPU temperatures. However, these benefits come with trade-offs, as increased viscosity in nanofluids leads to higher pumping costs, and few studies have systematically addressed this balance between performance and economic feasibility. This study fills this gap by examining optimal nanofluid concentrations in immersion cooling for data centers, aiming to identify the most cost-effective balance of thermal performance and energy costs. Through a combined thermal and economic analysis, this work provides practical insights into optimizing nanofluid-based cooling systems for large-scale applications.

Within this study the thermal efficiency and flow characteristics of a data center server are examined with economic analysis. Some common nanofluids as a coolant, including Al_2O_3 , TiO_2 and CuO nanoparticles (0 %–4 %) are added in this study based on the Iceotope server model used. The Iceotope server is simulated using commercial CFD, COMSOL Multiphysics® software for conjugate convection. The second part of this study is considered to find the optimum volume fraction in the economic analysis, which corresponds to the estimated overall cost of a nanofluid in the data centre. The ultimate goal is to determine whether nanofluids can offer a practical and economically viable solution for enhancing performance in data centre servers.

2. Numerical analysis of immersed server cooling

2.1. Geometry and materials properties

The immediate contact liquid cooling system is used in two main ways; the servers may be immersed in either an open or a closed liquid dielectric bath. Fig. 1 clearly shows another immersion technique offered by Iceotope, the manufacturer of liquid cooling equipment. The server motherboard is immersed in a bath filled with dielectric liquid hydrofluoroethers (HFE) by adding heat exchanger (cooling plate) on the server bath wall to dissipate the heat produced by the electrical components, the cooling is done by circulating coolant (water) through the cooling plate.

The server's design follows the geometry of the Iceotope server, featuring dimensions of 25 cm in width, 40 cm in length, and 3 cm in height, and accommodating two CPUs. The cooling system analyzed is a serpentine tube configuration that cools both CPUs by circulating a coolant efficiently. This design maximizes heat dissipation by ensuring the serpentine tubes cover the entire surface area of the CPUs. The CPU (chip) has 160 W / m.K of aluminum thermal conductivity with dimensions of $5 \times 5 \times 0.5$ cm (wide x long x high), which positioned opposite to the cooling plate. The chip's heat sink fins are copper with 400 W / m. K thermal conductivity, and its geometry is 9.0 cm, 10 cm and 0.6 cm (wide, long and high, respectively). The heat sink contains 23 fins 23 and each fin has a width of 0.2 cm, a length of 10 cm and a height of 0.6 cm, which occupy approximately 51 % of the total volume. These fins serve as thermal conductivity enhancers, aiding in effective heat dissipation from the CPU to the cooling medium. The cooling plate is made of aluminium, having similar thermal properties as the chip with a rectangular configuration (25 cm wide, 40 cm long and 1.5 cm high) on the server's top surface. the plate has a circular tube along with the 1.0 cm diameter. A schematic representation of this cooling configuration is provided in Fig. 2.

Fig. 2 illustrates the server equipped with cooling plates containing two CPUs. Both CPUs share identical geometry and boundary conditions, as they are the primary components absorbing power during operation [34]. The server structure is divided into two parts: the first encloses the CPU (chip) immersed in a dielectric liquid (HEF) that dissipates heat (q) through natural convection. The second part, serving as a cooling module, is a cooling plate with the serpentine tubes to dissipate the heat of the sealed enclosure using chilled nanofluid or water at a low inlet temperature (T_c). The server model is aligned perpendicularly and

is assumed to be externally insulated. The flow through the serpentine tube bends in the cooling plates is laminar, given that the fluid velocity required to maintain the CPU at a stable operating temperature results in a Reynolds number (Re) of <2300 .

Using the Re number equation with the assumption that all power consumption of the processors (CPU) is dissipated, as the chip's heat and secure working temperature (CPU) is confined to less than $< 85^\circ\text{C}$ [35]. The power usage of the CPU will therefore be 80 W in the present study and the related Re number values for maintaining the Central Processing Unit (CPU) chip at a stable operating temperature vary from 200 to 700 for the serpentine tube cooling plate. As per the CPU's dissipated heat in the enclosure, turbulent flow of the dielectric liquid in the enclosure would be carried out since the calculated Rayleigh number ($Ra = gL^3\beta\Delta T/(\alpha\nu)$) was about 10^8 and for $Ra \geq 10^6$ the flow will be turbulent [36]. Where β , α , ν , g , and L are coefficient of thermal expansion, thermal diffusivity, dielectric liquid kinematic viscosity (HFE), acceleration of gravity and characteristic length, respectively. While ΔT is the discrepancy in temperature between the hottest and coldest enclosure. Nanofluids have a lower Reynolds number compared to base fluids because of their comparatively low viscosity. A modification of the cooling plate mass flow rate ($m\&$) is then conducted to attain a specific base fluid Re number based on the Re number formula, $m\& = Re\mu A/D$, where A represents the cross-sectional area of the inlet cooling plate, and μ denotes the dynamic viscosity of both water and nanofluids. The diameter of the cooling plate tube is denoted by D .

In this study, the thermophysical properties of the base fluid (water) and nanofluids were extracted from reliable literature sources. For water, the properties were treated as temperature-dependent as outlined [37], while the properties of the nanoparticles Al_2O_3 , TiO_2 , and CuO at room temperature were obtained from [30]. These properties are detailed in Table 1.

The thermophysical properties of the nanofluids used in this study were calculated using established mixture theory and thermal equilibrium models. These approaches have demonstrated strong agreement with experimental data for various nanofluids. The density and specific heat of the nanofluids were computed using the following relationships:

$$\text{Nanofluid density } (\rho_{nf}) \text{ is determined as: } \rho_{nf} = \rho_{bf}(1 - \phi) + \rho_p\phi \quad (1)$$

$$\text{Nanofluid heat capacity } (C_{p,nf}): (\rho C_p)_{nf} = (\rho C_p)_{bf}(1 - \phi) + \phi(\rho C_p)_p \quad (2)$$

where the volume fraction, density and specific heat of the nanoparticle

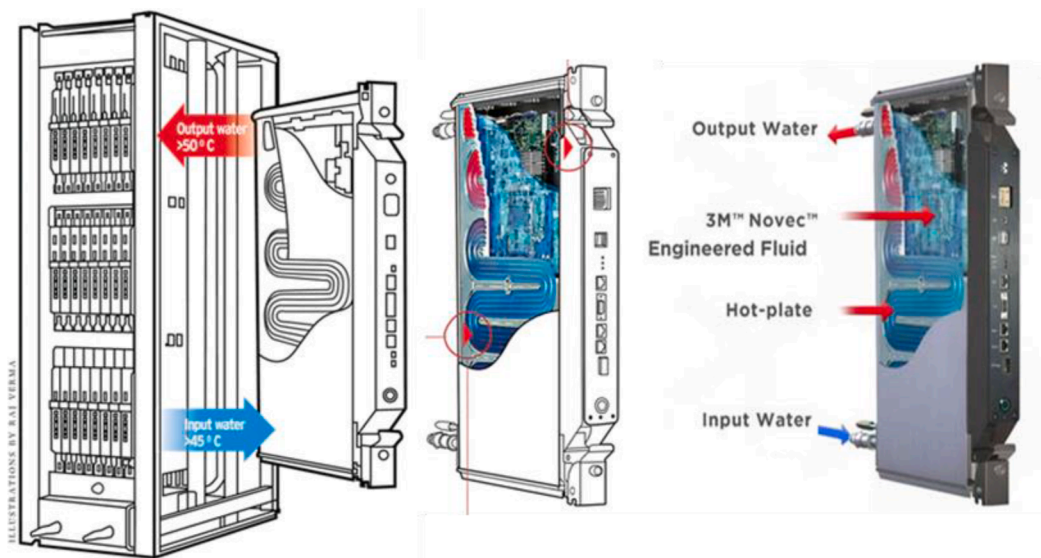


Fig. 1. Schematic of Iceotope's server [33].

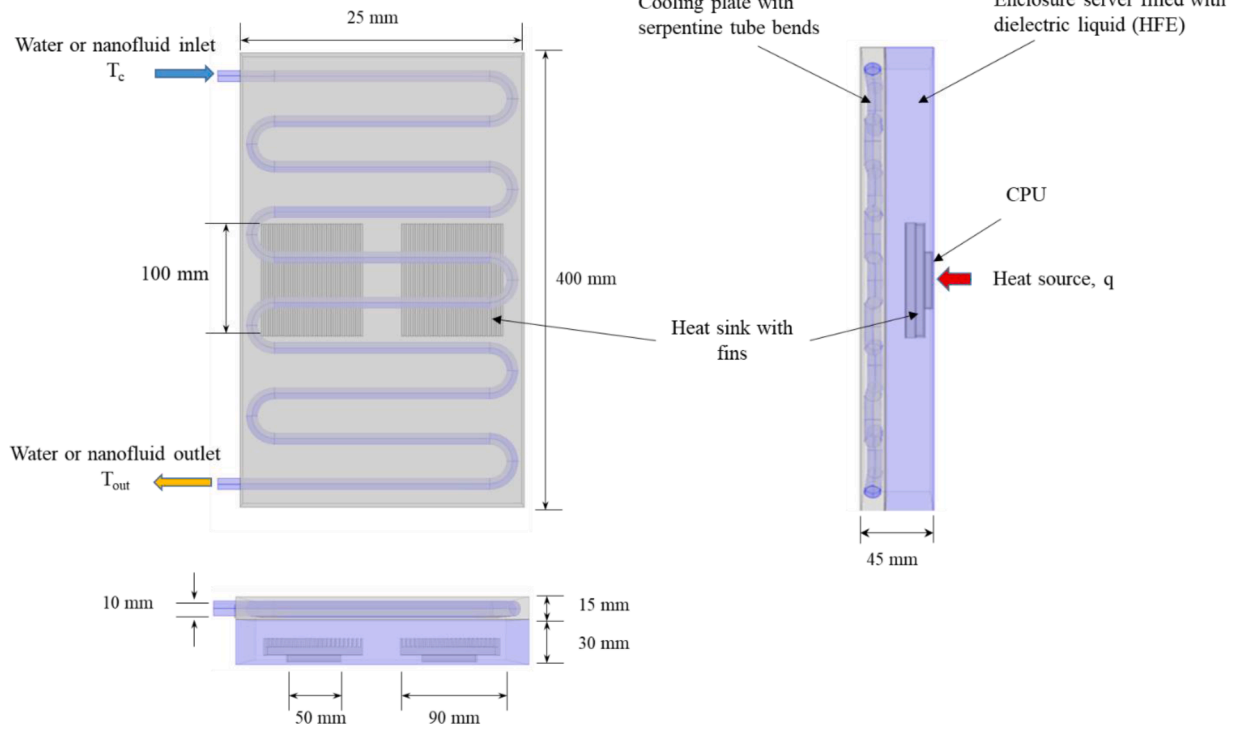


Fig. 2. Schematic showing the geometry and boundary conditions of the physical server model.

Table 1
Characteristics of nanoparticles used.

Type of nanoparticle	Units	Al ₂ O ₃	CuO	TiO ₂
Density of nanoparticle (ρ_p)	kg/m ³	4000	6300	4200
Thermal conductivity (k_p)	W/m.K	42	20	12
Heat capacity ($C_{p,p}$)	J/kg.K	890	550	720
Radius of nanoparticle	nm	50	30	45

are ϕ , ρ_p and $C_{p,p}$, respectively.

The thermal conductivity model proposed by Murugesan Chandrasekar et al. [38] is employed. This model, which incorporates particle shape, nano-layer thickness, and Brownian motion, is specifically applied to Al₂O₃/water, CuO/water, and TiO₂/water nanofluids. By accounting for these factors, the model offers a comprehensive and accurate representation of the thermal conductivity characteristics of each nanofluid, aligning well with their unique properties and enhancing the reliability of the analysis.

Nanofluids thermal conductivity (k_{nf}) can be written [38] as:

$$k_{nf} = k_{bf} \left(\frac{k_p + (n-1)k_{bf} + (n-1)(k_p - k_{bf})(1+\beta)^3\phi}{k_p + (n-1)k_{bf} - (k_p - k_{bf})(1+\beta)^3\phi} \right) \quad (3)$$

where k_{bf} and k_p represent the pure fluid and the nanoparticles thermal conductivities, respectively. While the parameter β denotes the ratio of the nanoparticle size to the fluid layer around the nanoparticle.

The dynamic viscosity of nanofluids was determined based on experimental data from previous studies, reflecting variations influenced by nanoparticle type, load, and containing fluid temperature [12] as detailed below:

$$\frac{(\mu_{nf})_{Al_2O_3}}{\mu_{bf}(T)} = \exp(6.599\phi / (0.288 - \phi)) \quad \text{for } \phi = 0 \text{ to } 4.0 \text{ vol.}\% \quad R^2 = 0.9973 \quad (4)$$

$$\frac{(\mu_{nf})_{TiO_2}}{\mu_{bf}(T)} = 25.17\phi^2 + 29.562\phi + 1 \quad \text{for } \phi = 0 \text{ to } 4.0 \text{ vol.}\% \quad R^2 = 0.9977 \quad (5)$$

$$\frac{(\mu_{nf})_{CuO}}{\mu_{bf}(T)} = 776.28\phi^2 + 7.7392\phi + 1 \quad \text{for } \phi = 0 \text{ to } 4.0 \text{ vol.}\% \quad R^2 = 0.9963 \quad (6)$$

The viscosity significantly impacts the pressure drop in the system, with higher viscosity nanofluids requiring increased pumping power to maintain flow. This increase in viscosity, particularly at higher nanoparticle concentrations, can reduce the cost-effectiveness of using nanofluids despite their enhanced thermal properties.

Regarding the thermo-physical properties of the dielectric liquid (HFE) within the server enclosure, they were provided by the supplier as shown in Table 2. The temperature-dependent properties of density and viscosity were calculated as functions of fluid temperature using empirical relationships as shown below:

2.2. Mathematical formulation

Two modes of heat transfer were observed within the server: forced

Table 2
Shows properties of the dielectric fluid dielectric liquid (HFE) [39].

Property	Dielectric liquid (HFE)
Density, ρ (kg/m ³)	1716 – 2.4T where T is liquid temperature in (°C)
Kinematic viscosity, ν (m ² /s)	$(2.745 \log(T) - 14.46)^{-2.17}$ where T is liquid temperature in (K)
Specific heat, C_p (J/kg.K)	1140
Thermal conductivity, k (W/m.K)	0.063
Thermal expansion coefficient, β (1/K)	0.00151

convection through the cooling plates tube and natural convection of the dielectric fluid (HFE) inside the server enclosure. To evaluate the heat transfer and fluid flow dynamics, the Navier-Stokes equations, energy equation, and continuity equation were solved. These governing equations were applied to a defined state, incorporating appropriate initial and boundary conditions for the system. The Reynolds number plays a key role in determining the flow regime of the nanofluid, with higher Reynolds numbers leading to turbulent flow, which enhances convective heat transfer and thus cooling performance. Optimizing the Reynolds number helps achieve an effective balance between cooling efficiency and pumping energy costs.

The following assumptions were made to simplify the mathematical modelling of the cooling system:

- **Steady-State Conditions:** The model assumes a steady-state flow, meaning that all properties, such as temperature and flow rates, remain constant over time.
- **Incompressible Flow:** The nanofluid is treated as incompressible, with constant density across the system, which is a standard assumption in thermal-fluid simulations.
- **No Chemical Reactions:** The model assumes that no chemical reactions occur between the nanoparticles and the base fluid.
- **Homogeneous Mixture:** The nanofluid is treated as a homogeneous mixture, assuming that nanoparticles are uniformly dispersed in the base fluid and behave as a single-phase fluid.
- **Negligible Nanoparticle-Base Fluid Heat Transfer:** The heat transfer between individual nanoparticles and the base fluid is assumed to be negligible, allowing us to model the nanofluid as a single entity with averaged thermophysical properties.
- **Laminar and Turbulent Flow Conditions:** Depending on the Reynolds number, the model accounts for both laminar and turbulent flow regimes within the cooling channels.

These assumptions provide a foundational basis for the modelling equations used in this study. They simplify the complex interactions within the cooling system while maintaining sufficient accuracy to predict the thermal and hydraulic behavior of the nanofluids.

2.2.1. Laminar flow inside serpentine tube bends

Based on Reynolds number calculations, the fluid flow in the cooling plate tubes was determined to be laminar. Regarding the behavior of nanofluids, it is assumed that there is no-slip between the pure water and nanoparticles. This assumption allows the combination of pure water and nanoparticles to be treated as a homogeneous nanofluid with altered thermophysical properties, resulting in the application of the governing equations presented earlier.

The flow and heat transfer calculations in the cooling plates are conducted under the following key assumptions: steady-state conditions, incompressible fluid, no chemical reactions, a dilute nanofluid mixture, negligible viscous dissipation, and negligible heat transfer between the nanoparticles and the base fluid (pure water). Given these assumptions, the general governing equations are simplified to the following forms:

$$\text{The continuity equation : } \nabla \cdot \mathbf{V} = 0 \quad (7)$$

$$\text{The equation of momentum : } \rho(\mathbf{V} \cdot \nabla) \mathbf{V} = \rho \mathbf{g} - \nabla P + \nabla \cdot \boldsymbol{\mu} \nabla \mathbf{V} \quad (8)$$

$$\text{The equation of energy : } (\rho C_p) \mathbf{V} \cdot \nabla T = \nabla \cdot (k \nabla T) \quad (9)$$

2.2.2. Turbulent flow inside enclosure

Turbulent flow is characterized by chaotic and fluctuating motion, with eddies of varying sizes and durations. In the case of natural convection, driven by buoyancy forces, the fluid within the enclosure moves as described previously. This fluid and heat transition is modeled by solving the governing equations and applying the Boussinesq approxi-

mation to account for gravity effects. Multiple approaches exist for resolving turbulent flow, and in this analysis, the Reynolds-Averaged Navier-Stokes (RANS) equations are employed, a widely accepted method in Computational Fluid Dynamics (CFD) models. The fundamental equations governing turbulent flow are expressed as follows:

$$\text{The continuity equation : } \nabla \cdot \mathbf{V} = 0 \quad (10)$$

$$\text{Equation of momentum : } \rho(\mathbf{V} \cdot \nabla) \mathbf{V} = -\nabla P + \nabla \cdot \boldsymbol{\mu} \nabla \mathbf{V} + \nabla \cdot \boldsymbol{\tau}_{\text{turbulent}} + \mathbf{F} \quad (11)$$

$$\text{Equation of energy : } (\rho C_p) \mathbf{V} \cdot \nabla T = \nabla \cdot (k \nabla T) + Q \quad (12)$$

The term $\boldsymbol{\tau}_{\text{turbulent}}$ on the right-hand side of Eq. (11) accounts for the fluctuating eddies in turbulent flow, represented by the specific Reynolds stress tensor. Several turbulence models are available for modeling such flows. The most commonly used models are the k - ε and k - ω . For this analysis, the k - ε model is utilized to resolve the turbulent terms, as developed by Launder and Spalding. In addition to solving the continuity, momentum, and energy equations, this model introduces two additional transport equations for the turbulent kinetic energy (k) and its dissipation rate (ε):

Turbulent kinetic energy equation:

$$\rho \frac{dk}{dt} + \rho \mathbf{u} \cdot \nabla k = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_k} \right) \nabla k \right] + p_k - \rho \varepsilon \quad (13)$$

Turbulent dissipation rate equation:

$$\rho \frac{d\varepsilon}{dt} + \rho \mathbf{V} \cdot \nabla \varepsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_{e1} \frac{\varepsilon}{k} p_k - C_{e2} \rho \frac{\varepsilon^2}{k} \quad (14)$$

The production term (p_k) is given by

$$p_k = \mu_T \left[\nabla \mathbf{V} : (\nabla \mathbf{V} + (\nabla \mathbf{V})^T) - \frac{2}{3} (\nabla \cdot \mathbf{V})^2 \right] - \frac{2}{3} \rho k \nabla \cdot \mathbf{V} \quad (15)$$

The turbulent dynamic viscosity (μ_T) is defined as

$$\mu_T = \rho C_\mu \frac{k^2}{\varepsilon} \quad (16)$$

In these equations, $\mathbf{V}$, $\mathbf{V}^T$, μ_T , ρ , σ_ε and σ_k represent the average velocity, turbulent velocity transport, turbulent viscosity, fluid density, and the respective Prandtl numbers for ε and k . The constant parameters for the empirical turbulence model are:

$$\sigma_k = 1, \sigma_\varepsilon = 1.3, C_{e1} = 1.44, C_\mu = 0.09 \text{ and } C_{e2} = 1.92$$

2.2.3. The conjugate heat transfer mode

In the server model, heat transfer occurs through both conduction and convection. Conduction dominates within the solid components, such as the heat sink and cooling plates, while convection governs the heat transfer between the nanofluid and solid surfaces. This combined heat transfer mechanism is referred to as conjugate heat transfer. The conduction heat transfer in solid materials is described by the general heat conduction equation. The heat flux between the solid and fluid at the interface is governed by Fourier's law:

$$\nabla k_s \nabla T_s = \nabla \cdot k_f \nabla T_f \quad (17)$$

Where T_s and T_f are the solid and fluid temperatures at the wall, respectively.

3. Economic analysis of the use of nanofluids

The primary objective of the economic analysis is to evaluate the feasibility of utilizing nanofluid coolants in data center immersion cooling systems. This analysis examines the costs and benefits of incorporating nanofluids into such systems by translating all cash flows over the project's lifespan into a present value figure. The cost components include the thermal efficiency gains and the increased pressure

drop associated with higher nanoparticle concentrations, both of which contribute to the overall operational cost of the cooling system.

The relationship between nanofluid concentration and operational costs stems from two main factors: the cost of the nanofluid, which increases with the concentration of nanoparticles, and the energy savings resulting from reduced CPU temperatures. Nanofluids enhance heat transfer efficiency, which lowers the CPU's operating temperature and, consequently, decreases energy consumption. However, the higher viscosity of nanofluids, particularly at higher concentrations, increases the pumping power required to circulate the fluid through the cooling system. Therefore, the optimal nanofluid concentration is the one that minimizes the combined cost of nanofluid purchase and energy consumption (including pumping power, server energy, and refrigeration costs) over the system's lifespan.

3.1. Economic model framework

The economic model used in this analysis is based on the present worth method, which is widely applied in engineering projects to consolidate future cash flows into a single present value. This approach considers the initial investment for nanofluid acquisition and the recurring annual costs of energy consumption. The total cost (C_{tot}) is calculated using the following formula:

$$C_{tot} = C_{nf}(\phi) + C_e \text{ PWF} \quad (18)$$

Where C_{nf} represents the initial cost of purchasing the nanofluid, which depends on the nanoparticle concentration (ϕ), C_e the total annual energy cost, and PWF is the present worth factor.

The present worth factor (PWF) is derived from the inflation rate (r_i), discount rate (r_d), and the system's lifespan (m years), following standard economic analysis formulas. This formula allows for the comparison of cash flows under different scenarios of inflation and interest rates.

$$PWF = \left(\frac{1 + r_i}{r_d - r_i} \right) \left[1 - \left(\frac{1 + r_i}{1 + r_d} \right)^m \right] \quad (19)$$

2.2. Energy consumption and cost analysis

The total annual energy consumption (E_{tot}) includes the power used by the CPU (E_s), the cooling system (E_c), and the pump (E_p). As the concentration of nanoparticles increases, the CPU's operating temperature decreases, reducing the power required to run the processor. However, the energy required to pump the nanofluid increases due to the higher viscosity of the fluid, especially at higher nanoparticle concentrations.

The energy required to cool the system using vapor-compression refrigeration is given by:

$$E_c = Q_g / p_c \quad (20)$$

where Q_g is the heat generated by the CPU, and p_c is the coefficient of performance of the refrigeration system, which typically ranges from 2 to 12, depending on the system's efficiency. In this study, a p_c value of 5 is assumed.

The total CPU energy consumption (E_s) is a combination of static and dynamic power components, expressed as:

$$E_s = P_{static} + P_{dynamic} \quad (21)$$

Dynamic power ($P_{dynamic}$) is modeled as a function of the processor's activity factor (α), voltage (V), and clock speed (ω):

$$P_{dynamic} = \alpha \omega V^2 \quad (22)$$

Static power (P_{static}) is related to the leakage current, which varies with temperature and voltage. The leakage current is described by the Frenkel-Poole emission model, and the total energy consumption is

adjusted based on CPU temperature.

$$P_{static} = BV^2 \exp\left(-\frac{E - C\sqrt{V}}{T}\right) \quad (23)$$

Where B , C and E are constants associated with the CPU and its operation, and where T is the CPU temperature (K).

The CPU energy consumption (E_s) can be extracted from as follow:

$$E_s = BV^2 \exp\left(-\frac{E - C\sqrt{V}}{T}\right) + \alpha \omega V^2 \quad (24)$$

The processor (CPU) selected for this study is configured with the following parameter values:

$$\alpha = 14.91, B = 4408.2, C = 56.4, E = 2020.9$$

The energy consumption of the CPU is determined by substituting these values into the relevant equations.

$$E_s = 4408.2V^2 \exp\left(-\frac{2020.9 - 56.4\sqrt{V}}{T}\right) + 14.91\omega V^2 \quad (25)$$

The operational parameters of the CPU, including voltage (V) and clock speed (ω), were adjusted to produce 80 W of heat under standard operating conditions with the base fluid at a lower Reynolds number. Specifically, by applying Eq. (25), at $Re = 200$ and $V = 1.124$ V, the clock speed of the CPU is set to 3.06 GHz to achieve the desired power output of 80 W.

As nanoparticle concentration increases, the pressure drop across the cooling system rises, necessitating higher pumping power to maintain adequate fluid circulation. The energy consumption of the pump (E_p) is calculated as:

$$E_p = \frac{m\dot{\Delta}p}{\rho p_p} \quad (26)$$

Where $m\dot{\Delta}$ is the mass flow rate, Δp is the pressure drop, ρ is the fluid density, and p_p is the pump efficiency, assumed to be 0.85 in this study. The increased pressure drop is a function of both fluid viscosity and flow rate, and it directly affects the total energy costs of the system.

The total cost optimization is driven by two opposing factors: reducing CPU operating temperatures and increasing pumping power requirements. As the nanoparticle concentration increases, thermal performance improves, but the associated pressure drop and energy consumption also increase. Therefore, an optimal nanoparticle concentration exists at which the overall system cost is minimized.

In the current analysis, the optimal nanofluid concentration is determined by minimizing the total cost (C_{tot}) over the system's lifespan, balancing the initial cost of the nanofluid with the energy savings from improved thermal performance and the increased costs of pumping.

The total annual energy consumption of the cooling system is represented by:

$$E_{tot} = E_c + E_p + E_s \quad (27)$$

Where E_c , E_p , and E_s correspond to the energy used by the cooling system, pump, and CPU, respectively, as mentioned before. Consequently, the annual energy cost (C_e) is calculated as:

$$C_e = E_{tot} C_s \quad (28)$$

Here, C_e represents the cost of electricity per kWh in the local market. Typically, the total heat load generated by the CPU (Q_g) and the pressure drop associated with different nanofluid volume fractions (ϕ) are the key factors influencing the overall energy cost. These values are derived from the numerical simulations discussed in the previous section, with the corresponding parameter values presented in Table 3 [40–45].

For this analysis, it is assumed that the server operates continuously (24 h a day) throughout its lifespan. The volume of nanofluid or base fluid utilized in the cooling plate is derived from the total cooling capacity, which is estimated at 0.25 liters for the system. Table 4 lists the

Table 3

Nominal values of the factors used in the model for optimisation.

Items	Nominal values	Comments
C_{nf} (\$/kg)	Cost depends on the type and concentration of the nanofluid used	Varies by nanofluid and concentration
c_e (\$/kWh)	0.145	Cost of electricity
P_c	5	Coefficient of performance (COP) for the cooling system
P_p	0.85	Pump performance factor
m (years)	3	Lifespan of the system
r_d (%)	3.4	Discount rate
r_i (%)	2.7	Inflation rate

Table 4

Nanofluid cost in commercial companies (Alfa Company) [46].

Nanofluids	c_{nf} (\$/kg)**	Comments
Al ₂ O ₃ /water	32.39	For concentration 6.0 vol %
CuO/water	597.64	For concentration 13.6 vol%
TiO ₂ /water	90.45	For concentration 11.0 vol%

** Nanofluid cost at concentration purchase.

costs of nanofluids based on local market prices at the time of the study. The price of nanofluids depends on several factors, including the type of base fluid (e.g., water), nanoparticle concentration, particle size, purity, and the method of preparation. The cost of nanofluids can be calculated based on mass fraction, as determined by commercial costs in dollars per kilogram (\$/kg). For instance, the price of Al₂O₃-water nanofluid with a 6.0 vol % concentration is \$32.39/kg. The relationship between nanofluid cost and nanoparticle concentration is approximately linear, taking into account the cost of pure water (without nanoparticles), which is approximately \$0.47/kg at room temperature.

This analysis does not account for any specific exemptions or variations in the economic model. The system's chip is assumed to generate 80 W of power, and the temperature of the chip is derived accordingly. The processor's power consumption, which is a function of the chip temperature, is used in the economic model to determine energy usage. For this analysis, it is assumed that the cooling system's power consumption remains constant when using pure water as the cooling fluid. The cumulative cost includes both the cost of nanofluids as a function of nanoparticle concentration and the energy consumption over the lifespan of the system. Several factors influence the optimization process to determine the ideal nanoparticle concentration, such as CPU power consumption, the system's ability to cool 160 W of heat generated by the chips, and the pumping power required to overcome the pressure drop in circulating the cooling fluid throughout the server.

4. Results and discussions

This section evaluates the thermal and flow properties of server models utilizing various forms and concentrations of nanofluids at specific Reynolds numbers. Key attributes, such as the average peak surface temperature of chips and the pressure drop across the server cooling plate through the serpentine tube cooling system, have been previously measured and reported. These findings will be integrated into the subsequent section for economic analysis.

To ensure the reliability of the computational model, a validation comparison is conducted with the study by Raisi et al. [47], which examined forced convection in nanofluids across various volume fractions and Reynolds numbers. Specifically, the calculated Nusselt number and pressure drop for CuO-water nanofluid at a Reynolds number of 500 are compared as shown in Fig. 3. Results show close alignment with those reported by Raisi et al., with deviations of <3 % for the Nusselt number and 5 % for the pressure drop, indicating strong agreement. This consistency validates the model's accuracy in predicting heat transfer and hydraulic behavior in similar cooling systems, supporting its application for the present study.

The influence of the number of mesh elements on thermal efficiency was assessed through two key indicators: the average surface temperature of the chips and the pressure drop across the cooling plate at $Re =$

Table 5

The effect of the number of elements on average chip temperature and a decrease in the pressure of the plate cooling for water at $Re = 200$.

Numbers of element	Temperature (°C)	Pressure drop (Pa)
1,384,414	83.7	1.96
3,354,676	82.5	1.83
3,905,902	81.9	1.89
5,106,884	81.80	1.88

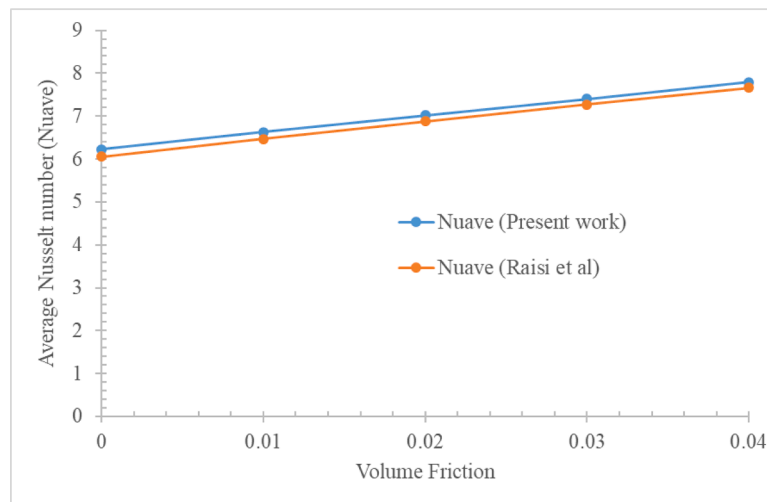


Fig. 3. Validation of Average Nusselt Number at $Re = 500$ for Different Volume Fractions, Comparing with Raisi et al.

200. Table 5 demonstrates that grid independence is achieved across all tested element counts. For element counts exceeding 3905,902, the error in monitored values (temperature and pressure drop) remains below 0.5 %. Accordingly, this number of elements was selected as the optimal choice for numerical simulations.

Based on Reynolds number calculations to maintain optimal operating temperatures, the flow through the serpentine tubes must remain within a range of 200 to 700. It is essential to highlight that the cooling plate achieves greater thermal efficiency in parallel flow configurations than in counter flow, particularly when the inlet flow enters from the top and exits at the bottom of the server. In parallel flow configurations, the cold liquid effectively surrounds the CPU (chip), which is immersed in a dielectric liquid (HEF) within an enclosure that relies on natural convection for fluid movement. This arrangement allows the warmed liquid at the bottom to rise, promoting circulation and reducing stagnant zones due to buoyancy-driven convection. In contrast, with counter flow configurations, the cold liquid enters from the bottom and flows upward, causing cold liquid to accumulate and remain stagnant around the CPU, thereby limiting natural convection. This stagnation reduces overall thermal efficiency as the enclosure's natural convection is inhibited. For these reasons, this study focuses on servers with parallel flow configurations (see Figs. 4(a), (4b) and Fig. 5) to maximize thermal efficiency by leveraging natural convection and minimizing stagnant regions.

4.1. Thermal and hydraulic performance of immersion cooling in data center servers

Following the mesh independence test and validation, this section examines the performance of the cooling system using various nanofluids within a server equipped with a cooling plate. Fig. 6 through 8 display the results for server chip temperature and pressure drop across the cooling plate. Three nanofluids Al_2O_3 -water, TiO_2 -water and CuO -water were applied at Reynolds numbers ranging from 200 to 700. The figures also demonstrate the impact of these nanofluids on the average surface temperature of the top chip within an enclosure cooled

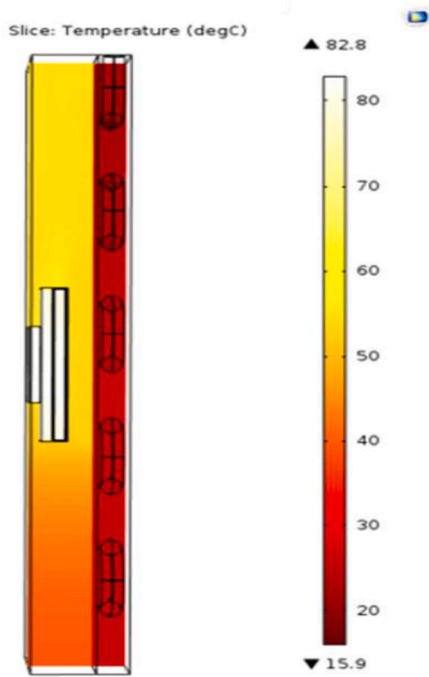


Fig. 4a. : Temperature of the cross section in the middle of the server; using water at Reynolds numbers $Re = 200$ parallel flow configuration enters from the top and exits at the bottom of the server.

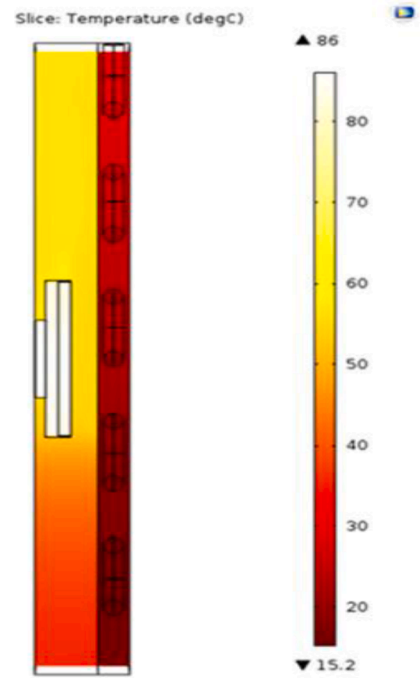


Fig. 4b. Temperature of the cross section in the middle of the server; using water at Reynolds numbers $Re = 200$ counter flow configuration enters from the bottom and exits at the top of the server.

by dielectric fluid (HFE). Furthermore, the pressure drop across the serpentine tube with each nanofluid was analyzed to provide insights into the cooling performance under different flow conditions.

Figs. 6–8 reveal a consistent decrease in chip temperature with increasing Reynolds number and volume fraction for each nanofluid. Generally, for a constant Reynolds number, the chip temperature declines linearly as the volume fraction increases. However, as the Reynolds number further increases, this temperature reduction becomes less significant. The temperature differential between low and high concentrations diminishes as Reynolds numbers rise; for example, at $Re = 200$, the temperature difference between 0 and 4 vol.% Al_2O_3 -water nanofluid is approximately 3 °C, while at $Re = 700$, the difference narrows to 1.25 °C.

Conversely, the pressure drop exhibits an opposite trend, increasing monotonically with both Reynolds number and volume fraction, as seen in Figs. 5–7. The pressure drop for specific Reynolds numbers grows substantially with increasing nanofluid volume fraction. For instance, the pressure drop for Al_2O_3 -water nanofluid increases by about 30 % at 0.5 vol.% and by about 600 % at 4 vol.% compared to the base fluid. Similar behavior is observed for TiO_2 -water and CuO -water nanofluids in Figs. 7 and 8, respectively.

Among the nanofluids, Al_2O_3 -water nanofluid demonstrates superior thermal performance due to its higher thermal conductivity, resulting in lower average chip temperatures compared to TiO_2 -water and CuO -water nanofluids. However, Al_2O_3 -water nanofluid also exhibits higher viscosity, leading to a greater pressure drop across the cooling plate. In cooling applications, maintaining a safe operating temperature for the chip is crucial for reliability. Thus, there is often a trade-off between reducing chip temperature and managing the associated increase in pressure drop.

In summary, Al_2O_3 -water nanofluids provide the lowest average chip temperatures, making them advantageous for improving chip reliability and energy efficiency [5,48–50]. However, the higher pressure drop, due to increased viscosity, results in higher pumping energy consumption as the volume fraction rises. This trade-off underscores the need for an economic analysis to determine the most cost-effective nanofluid that

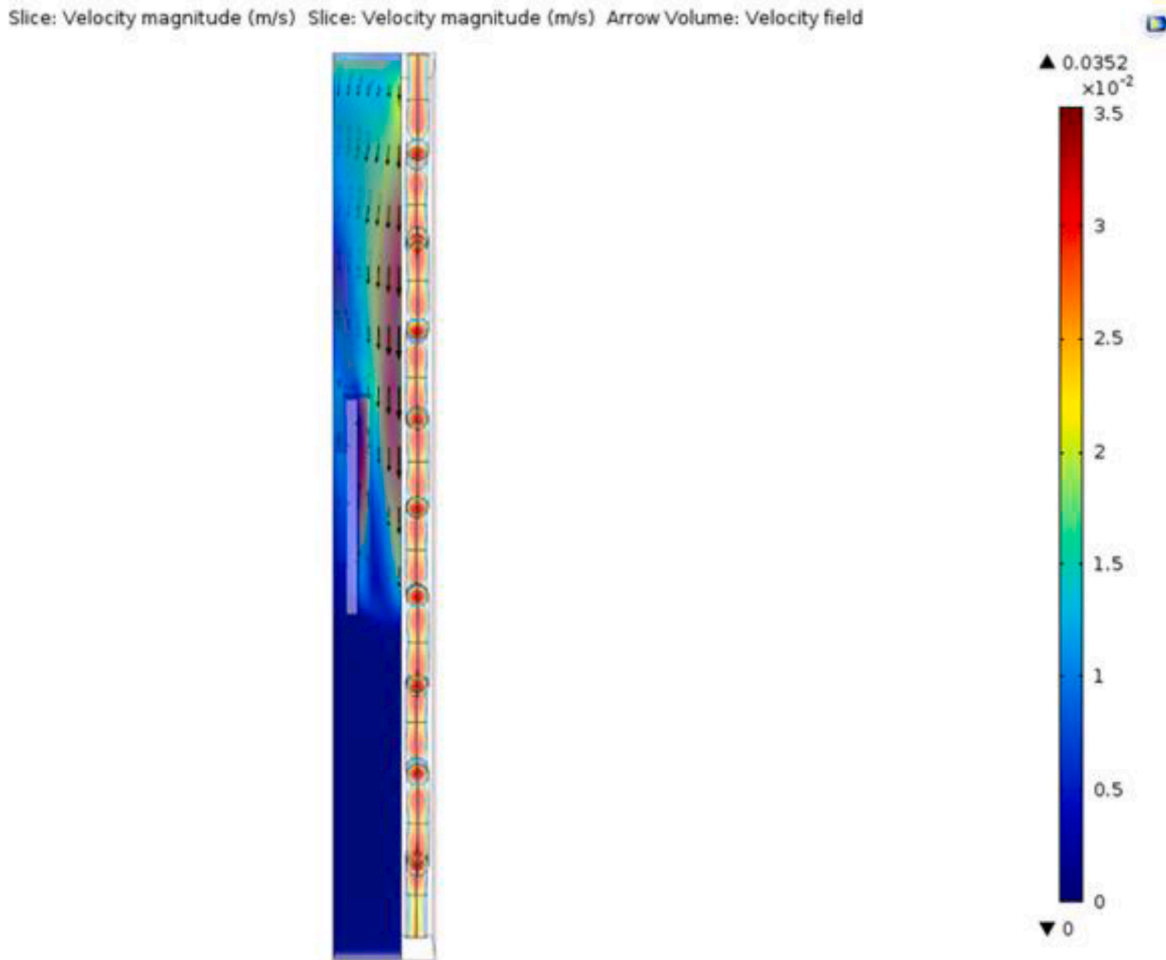


Fig. 5. Velocity field and arrow volume direction flow of the liquid dielectric bath and water in the cooled plate with the serpentine tube at $Re = 200$; showing parallel flow configuration enters from the top and exits at the bottom of the server.

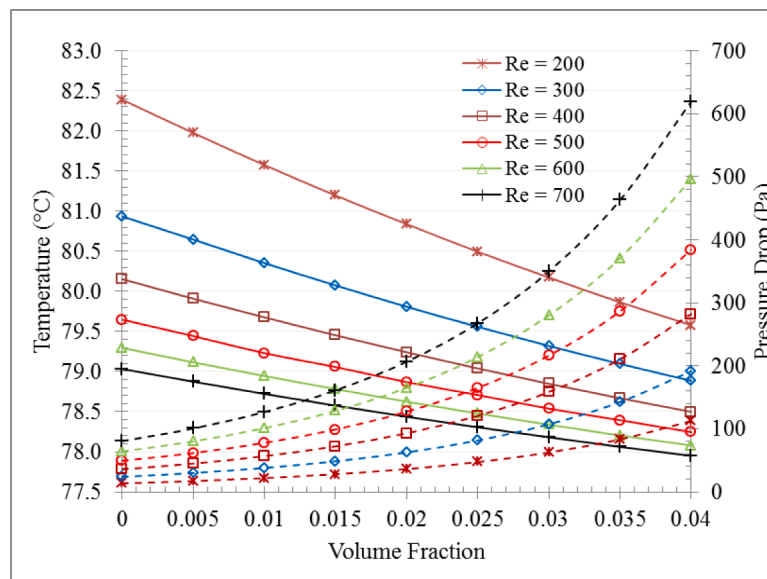


Fig. 6. Solid line (chip temperature) and dashed line (pressure drop); using Al_2O_3 -water nanofluids with different volume fractions and Re numbers.

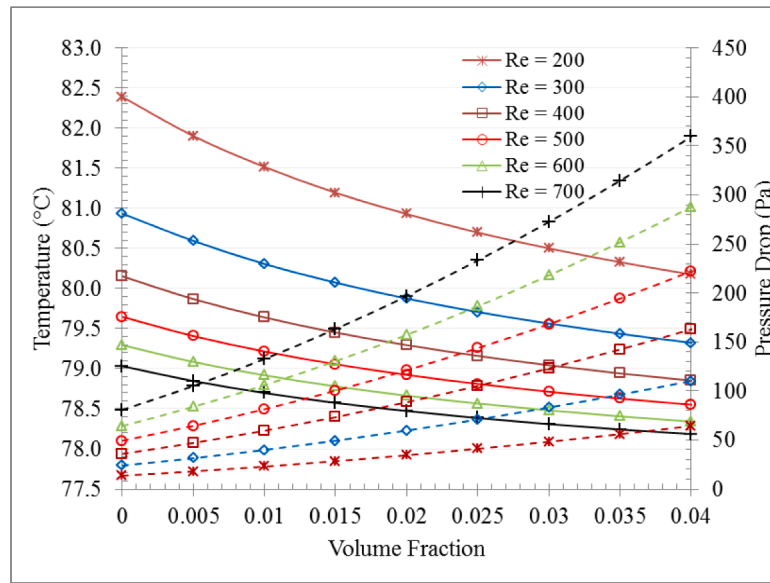


Fig. 7. Solid line (chip temperature) and dashed line (pressure drop); using TiO_2 -water nanofluids with different volume fractions and Re numbers.

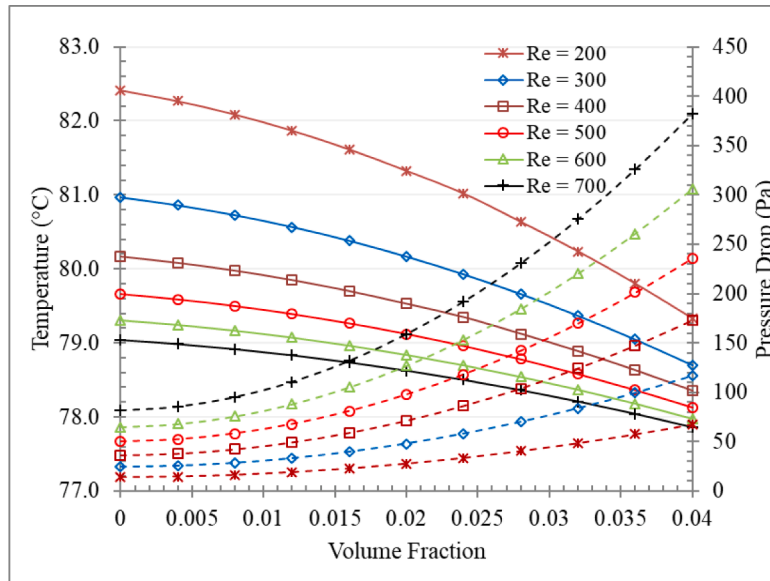


Fig. 8. Solid line (Chip temperature) and dashed line (pressure drop); using CuO -water nanofluids with different volume fractions and Re numbers.

can maintain an optimal chip temperature while minimizing the required pumping power.

4.2. Analysis of cost-effectiveness of nanofluid in immersion cooling in data center servers

Initially, Fig. 9 presents the performance of the base fluid under varying Reynolds numbers within the server model design. At lower Reynolds numbers, the total cost remains relatively high due to insufficient cooling of the CPU, which operates at elevated temperatures. As the Reynolds number increases, enhanced coolant circulation leads to a significant reduction in CPU temperature, resulting in a corresponding decrease in the total system cost. The analysis reveals that the optimal total cost, measured at \$706.98, is achieved at $Re = 600$, where the cooling efficiency is maximized, and the CPU operates at a stable, lower temperature. Beyond this point, however, further increases in Reynolds number result in diminishing returns, as the energy requirements for

pumping power begin to outweigh the cooling benefits. The rising cost of the pumping power offsets the savings from the reduced CPU temperature, rendering the system less economical at $Re > 600$. This demonstrates the trade-off between cooling efficiency and power consumption, emphasizing the importance of optimizing the Reynolds number for cost-effective operation.

Figs. 10–12 illustrate the variation in total cost as a function of volume fractions for the server model using Al_2O_3 -water, TiO_2 -water, and CuO -water nanofluids, respectively. Each curve identifies a minimum total cost, which corresponds to the recommended concentration for a specific Reynolds number. Notably, using the base fluid (water) at $Re = 600$ yields the lowest total cost, offering the lowest minimum total cost and the greatest cost effectiveness at \$706.98 under the conditions considered in this study.

In Fig. 10, the total cost variation using Al_2O_3 -water nanofluids is presented. The data show that at $Re = 200$, the minimum total cost is achieved at a concentration of 1.5 vol.%. The total cost curve changes

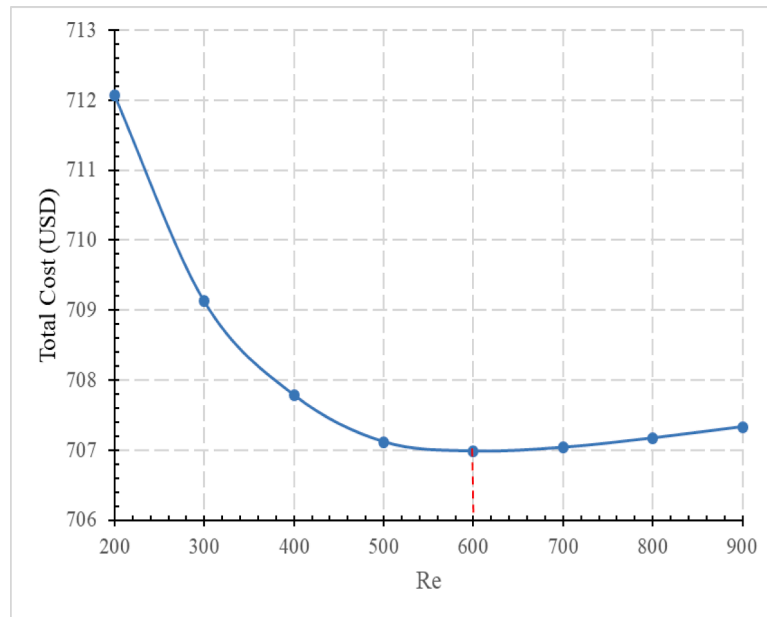


Fig. 9. Total cost variation with various Reynolds numbers using the base fluid (water).

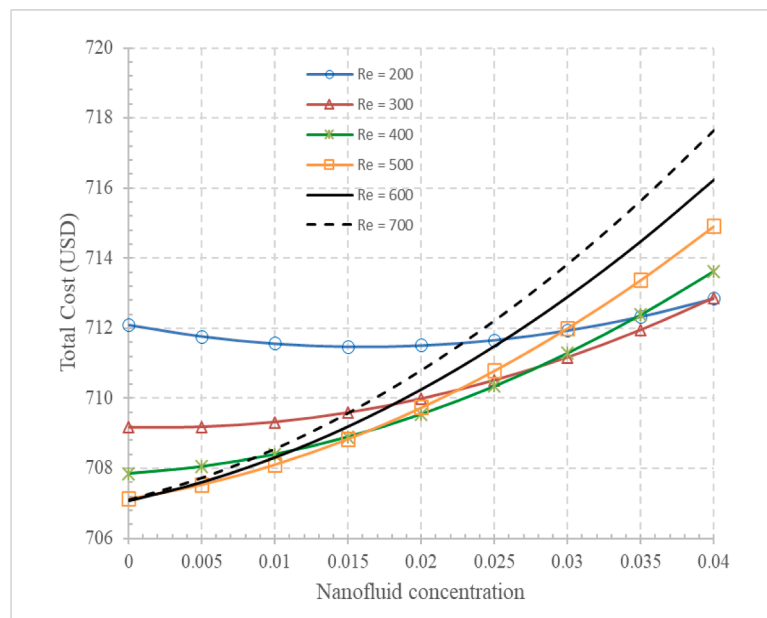


Fig. 10. Total cost difference for various volume fraction using Al₂O₃-water nanofluids.

more gradually between 1.0 vol.% and 2.0 vol.%, reflecting the stability of cost performance over this range. For other Reynolds numbers, the base fluid consistently provides the lowest cost, likely due to the added expense of nanofluid and increased pumping power. When examining TiO₂-water nanofluids (Fig. 11), the base fluid again results in the lowest total cost across all Reynolds numbers, except for $Re = 200$, where an optimum volume fraction of 0.5 vol.% for TiO₂-water is observed. Interestingly, at the same Reynolds number, Al₂O₃-water nanofluids exhibit a higher optimum volume fraction of 1.5 vol.%, along with a lower total cost. This difference can be attributed to the superior thermal conductivity and lower cost of Al₂O₃-water compared to TiO₂-water nanofluids.

Fig. 12 highlights the results for CuO-water nanofluids. Across all Reynolds numbers, the base fluid consistently delivers the lowest minimum total cost, and no nanofluid yields an optimum volume fraction.

Additionally, the total cost curve for CuO-water shifts sharply with small changes in volume fraction, a consequence of the high cost per unit of CuO-water and its relatively low thermal conductivity compared to Al₂O₃-water nanofluids. While nanofluids offer superior thermal performance, water is more cost-effective due to its low viscosity, which minimizes pumping energy requirements. Additionally, water's availability and low cost make it an economically viable option for large-scale applications where operational costs are a major consideration.

In terms of cost-effectiveness, the optimization is primarily driven by the cost and thermophysical properties of the nanofluids, including thermal conductivity and viscosity. It is important to emphasize that any variation in thermal properties or material costs for the different nanofluids would lead to significant changes in the optimization outcomes. Additionally, this optimization is highly application-dependent. Other critical properties, such as erosion and corrosion, which were not

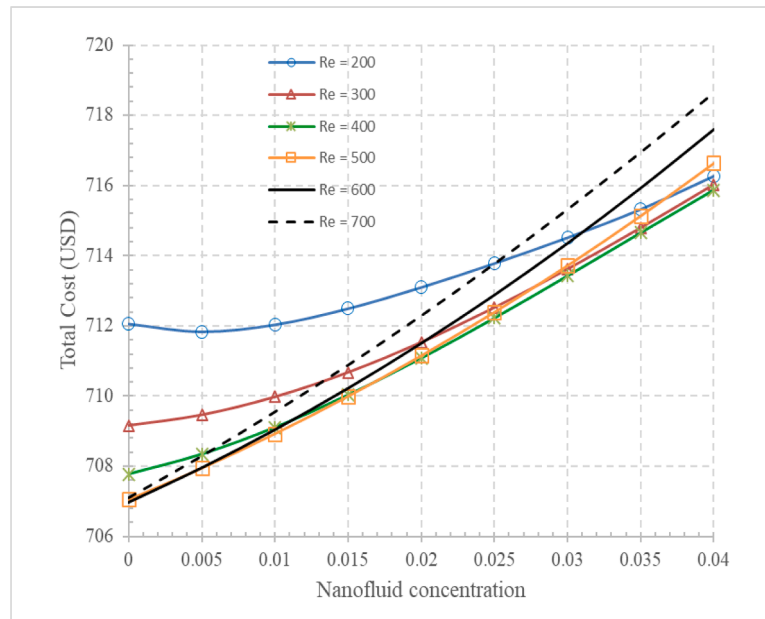


Fig. 11. Total cost difference with volume fraction using TiO_2 -water nanofluids.

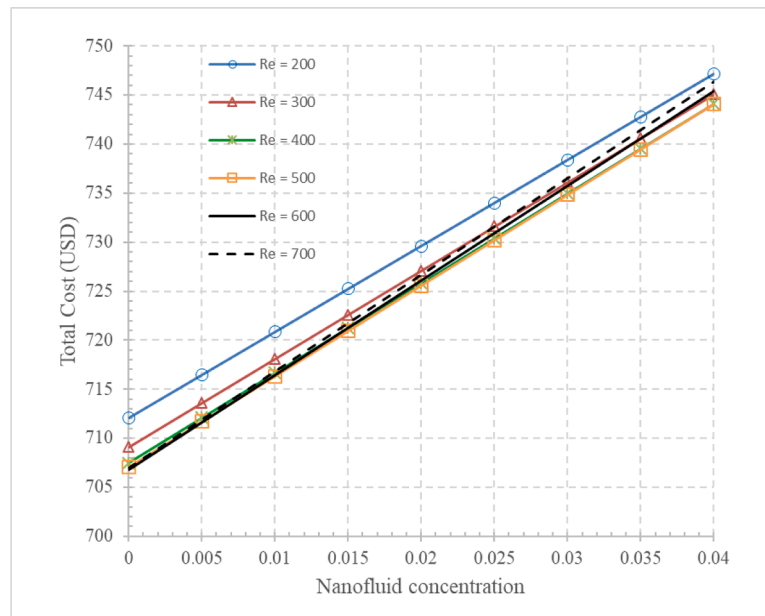


Fig. 12. Total cost difference with volume fraction using CuO -water nanofluids.

considered in this analysis, could play a pivotal role [51].

It is also noteworthy that increasing the Reynolds number to utilize nanofluids instead of the base fluid has practical limitations depending on the application configuration and the type and concentration of the nanofluid. For instance, increasing the Reynolds number to 200 for Al_2O_3 -water and TiO_2 -water nanofluids proves beneficial under the current conditions, whereas using CuO -water nanofluids does not provide significant advantages in this study.

Furthermore, any additional installation costs will not affect the optimum nanofluid concentrations. These costs would only shift the total cost versus volume fraction curve vertically. Expanding the study to include multiple servers, as typically found in a data center rack, would similarly not affect the optimal nanofluid concentration. Therefore, additional costs not accounted for in this analysis may be factored into the total cost, including the minimum total cost. Future work should

explore the impact of other economic factors, as mentioned in Table 3, on both the optimal nanofluid volume fraction and total cost.

5. Conclusions

This study evaluated the thermal performance and cost-effectiveness of nanofluids Al_2O_3 -water, TiO_2 -water, and CuO -water in immersion cooling systems for data center servers. Through simulations conducted in COMSOL Multiphysics®, the effects of varying Reynolds numbers (200–700) and nanoparticle volume fractions (0–4 %) on heat transfer and fluid flow were analyzed. Al_2O_3 -water nanofluid demonstrated superior thermal performance due to its higher thermal conductivity, resulting in significant reductions in CPU temperature, which enhanced energy efficiency. However, this came at the cost of increased viscosity, leading to greater pressure drops and higher pumping power

requirements, particularly at higher nanoparticle concentrations. TiO₂-water and CuO-water nanofluids also improved heat transfer but were less effective compared to Al₂O₃-water. In particular, CuO-water showed the highest overall cost due to its lower thermal conductivity and higher viscosity.

The economic analysis showed that while nanofluids improve cooling performance, pure water was the most cost-effective solution at an optimal Reynolds number of 600. This balance between thermal efficiency and minimal energy consumption highlighted pure water as the practical choice for data center cooling applications.

In conclusion, one primary limitation observed in this study is that increasing nanoparticle concentration, while enhancing thermal performance, also leads to higher viscosity, resulting in increased pressure drops and elevated pumping costs. This trade-off indicates an optimal concentration level beyond which the thermal benefits of nanofluids no longer justify the added operational expenses. Although nanofluids offer significant thermal advantages, their associated pressure drops and higher operational costs pose challenges. Therefore, the balance between thermal efficiency and energy expenditure must be carefully considered to optimize cooling systems effectively. Future research should explore advancements in nanofluid design, such as hybrid nanofluids and viscosity-controlling additives, as well as alternative cooling configurations to enhance both thermal performance and economic viability in data centre applications. These developments could improve heat transfer efficiency while minimizing operational costs, making nanofluids a more practical choice for large-scale cooling systems.

CRediT authorship contribution statement

Fahad S. Alkasmoul: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmed M. Almogbel:** Resources, Funding acquisition, Data curation. **Muhammad W. Shahzad:** Resources, Methodology, Funding acquisition, Formal analysis. **Amer J. Al-damook:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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